

COMP 8347: Internet Applications and Distributed Systems
Summer 2026 Lab #4

PART 1: Edit *models.py* to update existing models and create new models

```
from django.db import models
import datetime
from django.contrib.auth.models import User
from django.utils import timezone
```

1. Add the following model.

a. *Member* with fields below:

```
class Member(User):
    STATUS_CHOICES = [
        (1, 'Regular member'),
        (2, 'Premium Member'),
        (3, 'Guest Member'),
    ]

    status = models.IntegerField(choices=STATUS_CHOICES, default=1)
    address = models.CharField(max_length=300)
    city = models.CharField(max_length=20)
    province = models.CharField(max_length=2, default='ON')
    last_renewal = models.DateField(default=timezone.now)
    auto_renew = models.BooleanField(default=True)
    borrowed_books = models.ManyToManyField(Book, blank=True)
```

2. Create new db table. See what happens after each step.

- a. **Tools → Run manage.py Task...** (opens a window where you can type *manage.py* commands)
- b. In *manage.py* window: Type **makemigrations myapp** in dialog box.
- c. Optional step: Check latest file in *migrations* dir (use **sqlmigrate myapp ****** in *manage.py*)
- d. In *manage.py* window: Type **migrate**

3. Update db tables.

a. Add a new model **Order** with fields:

- *books* (*ManyToManyField(Book)*)
 - indicates the *books* that were part of this *order*.
- *member* (*ForeignKey(Member)*)
 - indicates the *member* that placed the *order*
- *order_type* (*IntegerField*)
 - choices of valid values = {0,1}. The default value is **1**. The values indicate the type of order and are interpreted as: [(0, 'Purchase'), (1, 'Borrow')]. **HINT:** Use similar format as *status* field in **Member** model.
- *order_date* (*DateField*)
 - indicates date *order* was placed. The default value is current date (i.e. *timezone.now*)

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- b. Add a new required field *country* to **Publisher** model, with a default value of 'Canada'. This indicates the *country* where the headquarters of the *publisher* is located.
- c. Add a new 'optional' field *description* to **Book** model. This provides a description of the *book*. The field should be of type *TextField*.
- d. Make the field *address* in **Member** model 'optional'. This field indicates the physical *address* of the *member*.
- e. Set the default value of *city* field in **Member** model to 'Windsor'.

Run **makemigrations**, **sqlmigrate** and **migrate** again until there are no errors. What is the latest file in *migrations* dir? Open it and check its contents.

PART 2: Enter data through Admin interface

1. Update *admin.py* as follows:

```
from django.contrib import admin
from .models import Publisher, Book, Member, Order

# Register your models here.
admin.site.register(Publisher)
admin.site.register(Book)
admin.site.register(Member)
admin.site.register(Order)
```

2. Start your server (**Run** → **Run 'mysiteF25'**) and navigate to admin site (127.0.0.1:8000/admin).
3. Login using *superuser* name and password (from Lab #3).
4. Enter the information for each *publisher*, *book*, *member* and *order* as given in *lab4data.pdf* through the admin interface. **Note:** Some data was already entered in Lab 3.

How is the data being displayed? Would it be more useful to display additional information?

5. a. Write `__str__` methods for each model.
 - b. For the **Order** model, write a method `total_items(self)` that returns the total *books* in the *order*.
6. In **Python console** import Django then models from *models.py*, then write queries to obtain the following information. Verify if your query generates the correct answer using *lab4data.txt*.

```
import django

from myapp.models import Publisher, Book, Member, Order
```

- a. List all the *books* in the db.
- b. List all the *members* in the db.

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- c. List all the *orders* in the db.
- d. List all the *publishers* in the db.

PART 3: Practice writing queries

1. Write queries to do the following.

- a. List all *Members* whose last name is 'Kwon'
- b. List all *Publishers* with headquarters in 'USA'
- c. List all *Members* that live in 'Ottawa'.
- d. List all *Members* that live on an 'Avenue' and live in ON *province*.
- e. List all the *Members* that have borrowed the book 'The Night Circus'
- f. List the *Books* that cost more than \$40.00
- g. List the *Members* that do NOT live in *province* ON.
- h. List the *Orders* placed by a *client* whose *first_name* is 'Elena'.
- i. List the *Members* whose member status are 'Regular Member'.
- j. List the *Books* that have between **300** and **500** pages (inclusive) and belong to *category* 'Science&Tech'.
- k. Get the first name of the *Members* who has borrowed 2 books.
- l. List all *Books* that the *Member* with username 'Marcus' is currently *borrowing*.
- m. List all the *Members* who live in ON and have *auto_renew* enabled.
- n. List the *Books* that 'Leo' has *purchased*.
- o. List the *city* where the headquarters of the *publisher* of the *book* purchased by 'Elena' is located.